

The Abstraction Gap Between Mathematics and Physics: The Hidden Compromise in Dimensional Analysis Resolved

Fredrik Bierich 
`fredrik.bierich@gmail.com`

September 29, 2026
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Abstract

In this dataset we bridge the abstraction gap between mathematics and physics by adopting a geometric perspective of trigonometry. We formalize this by addressing the arc length of a circular sector and defining the circular functions axiomatically as direct functional mappings from the physical parameters of a circular sector – namely the radius and the arc length – to its orthogonal projections. Finally, we demonstrate the utility of this framework by modeling simple harmonic motion and uniform circular motion through these geometric functions.

0.1 Circular Sector

0.1.1 Arc Length

$$\begin{aligned}s &= \lim_{n \rightarrow \infty} 2^n s_n \\ s_n &= \sqrt{r^2 + r^2 - 2ra_n} \\ a_n &= \frac{r\sqrt{r^2 - a_{n-1}^2}}{\sqrt{r^2 + r^2 - 2ra_{n-1}}} \\ a_0 &= x, -r \leq x \leq r \\ s &= \lim_{n \rightarrow \infty} 2^n \sqrt{r^2 + r^2 - 2ra_n} \\ a_n &= \frac{r\sqrt{r^2 - a_{n-1}^2}}{\sqrt{r^2 + r^2 - 2ra_{n-1}}} \\ a_0 &= x, -r \leq x \leq r \\ s &= \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \\ a_n &= \frac{r\sqrt{r^2 - a_{n-1}^2}}{\sqrt{r^2 + r^2 - 2ra_{n-1}}} \\ a_0 &= x, -r \leq x \leq r \\ s &= \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \\ a_n &= \frac{r\sqrt{r^2 - a_{n-1}^2}}{\sqrt{2r^2 - 2ra_{n-1}}} \\ a_0 &= x, -r \leq x \leq r \\ s &= \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \\ a_n &= r \frac{\sqrt{r^2 - a_{n-1}^2}}{\sqrt{2r^2 - 2ra_{n-1}}} \\ a_0 &= x, -r \leq x \leq r \\ s &= \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \\ a_n &= r \sqrt{\frac{r^2 - a_{n-1}^2}{2r^2 - 2ra_{n-1}}} \\ a_0 &= x, -r \leq x \leq r \\ s &= \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \\ a_n &= r \sqrt{\frac{(r + a_{n-1})(r - a_{n-1})}{2r^2 - 2ra_{n-1}}} \\ a_0 &= x, -r \leq x \leq r \\ s &= \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \\ a_n &= r \sqrt{\frac{(r + a_{n-1})(r - a_{n-1})}{2r(r - a_{n-1})}} \\ a_0 &= x, -r \leq x \leq r\end{aligned}$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n}$$

$$a_n = r \sqrt{\frac{r + a_{n-1}}{2r}}$$

$$a_0 = x, -r \leq x \leq r$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n}$$

$$a_n = r \sqrt{\frac{2r(r + a_{n-1})}{(2r)^2}}$$

$$a_0 = x, -r \leq x \leq r$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n}$$

$$a_n = r \sqrt{\frac{2r^2 + 2ra_{n-1}}{(2r)^2}}$$

$$a_0 = x, -r \leq x \leq r$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n}$$

$$a_n = r \sqrt{\frac{\left(\sqrt{2r^2 + 2ra_{n-1}}\right)^2}{(2r)^2}}$$

$$a_0 = x, -r \leq x \leq r$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n}$$

$$a_n = r \sqrt{\left(\frac{\sqrt{2r^2 + 2ra_{n-1}}}{2r}\right)^2}$$

$$a_0 = x, -r \leq x \leq r$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n}$$

$$a_n = r \frac{\sqrt{2r^2 + 2ra_{n-1}}}{2r}$$

$$a_0 = x, -r \leq x \leq r$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n}$$

$$a_n = \frac{r \sqrt{2r^2 + 2ra_{n-1}}}{2r}$$

$$a_0 = x, -r \leq x \leq r$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n}$$

$$a_n = \frac{\sqrt{2r^2 + 2ra_{n-1}}}{2}$$

$$a_0 = x, -r \leq x \leq r$$

0.1.2 Area

$$A = \frac{rs}{2}$$

0.1.3 Circular Functions

0.1.3.1 Arccosine

$$\begin{aligned} \arccos: \{(r, x) \in \mathbb{R}^2 \mid r > 0, -r \leq x \leq r\} &\rightarrow \mathbb{R} \\ (r, x) &\mapsto s \end{aligned}$$

$$\begin{aligned} \arccos(r, x) &= \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \\ a_n &= \frac{\sqrt{2r^2 + 2ra_{n-1}}}{2} \\ a_0 &= x \end{aligned}$$

0.1.3.2 Cosine

$$\begin{aligned} \cos: \{(r, s) \in \mathbb{R}^2 \mid r > 0\} &\rightarrow \mathbb{R} \\ (r, s) &\mapsto x \end{aligned}$$

$$\begin{aligned} s &= \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \\ a_n &= \frac{\sqrt{2r^2 + 2ra_{n-1}}}{2} \\ a_0 &= \cos(r, s) \\ s &= \lim_{m \rightarrow \infty} 2^m \sqrt{2r^2 - 2ra_{m,m}} \\ a_{m,n} &= \frac{\sqrt{2r^2 + 2ra_{m,n-1}}}{2}, n \leq m \\ a_{m,0} &= \cos(r, s) \\ \cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ a_{m,n-1} &= \frac{\sqrt{2r^2 + 2ra_{m,n}}}{2}, n \leq m \\ 2^m \sqrt{2r^2 - 2ra_{m,0}} &= s \\ \cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ 2a_{m,n-1} &= 2 \frac{\sqrt{2r^2 + 2ra_{m,n}}}{2}, n \leq m \\ 2^m \sqrt{2r^2 - 2ra_{m,0}} &= s \\ \cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ 2a_{m,n-1} &= \frac{2\sqrt{2r^2 + 2ra_{m,n}}}{2}, n \leq m \\ 2^m \sqrt{2r^2 - 2ra_{m,0}} &= s \\ \cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ 2a_{m,n-1} &= \sqrt{2r^2 + 2ra_{m,n}}, n \leq m \\ 2^m \sqrt{2r^2 - 2ra_{m,0}} &= s \\ \cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ (2a_{m,n-1})^2 &= \left(\sqrt{2r^2 + 2ra_{m,n}} \right)^2, n \leq m \\ 2^m \sqrt{2r^2 - 2ra_{m,0}} &= s \end{aligned}$$

$$\begin{aligned}
& \cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \\
& 2^2 a_{m,n-1}^2 = \left(\sqrt{2r^2 + 2ra_{m,n}} \right)^2, n \leq m \\
& 2^m \sqrt{2r^2 - 2ra_{m,0}} = s \\
& \cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \\
& 4a_{m,n-1}^2 = \left(\sqrt{2r^2 + 2ra_{m,n}} \right)^2, n \leq m \\
& 2^m \sqrt{2r^2 - 2ra_{m,0}} = s \\
& \cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \\
& 4a_{m,n-1}^2 = 2r^2 + 2ra_{m,n}, n \leq m \\
& 2^m \sqrt{2r^2 - 2ra_{m,0}} = s \\
& \cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \\
& 4a_{m,n-1}^2 - 2r^2 = 2r^2 + 2ra_{m,n} - 2r^2, n \leq m \\
& 2^m \sqrt{2r^2 - 2ra_{m,0}} = s \\
& \cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \\
& 4a_{m,n-1}^2 - 2r^2 = 2ra_{m,n}, n \leq m \\
& 2^m \sqrt{2r^2 - 2ra_{m,0}} = s \\
& \cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \\
& \frac{4a_{m,n-1}^2 - 2r^2}{2r} = \frac{2ra_{m,n}}{2r}, n \leq m \\
& 2^m \sqrt{2r^2 - 2ra_{m,0}} = s \\
& \cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \\
& \frac{4a_{m,n-1}^2}{2r} - \frac{2r^2}{2r} = \frac{2ra_{m,n}}{2r}, n \leq m \\
& 2^m \sqrt{2r^2 - 2ra_{m,0}} = s \\
& \cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \\
& \frac{2a_{m,n-1}^2}{r} - \frac{2r^2}{2r} = \frac{2ra_{m,n}}{2r}, n \leq m \\
& 2^m \sqrt{2r^2 - 2ra_{m,0}} = s \\
& \cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \\
& \frac{2a_{m,n-1}^2}{r} - r = \frac{2ra_{m,n}}{2r}, n \leq m \\
& 2^m \sqrt{2r^2 - 2ra_{m,0}} = s \\
& \cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \\
& \frac{2a_{m,n-1}^2}{r} - r = a_{m,n}, n \leq m \\
& 2^m \sqrt{2r^2 - 2ra_{m,0}} = s \\
& \cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m} \\
& a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r, n \leq m \\
& 2^m \sqrt{2r^2 - 2ra_{m,0}} = s
\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ a_{m,n} &= \frac{2a_{m,n-1}^2}{r} - r, n \leq m \\ \frac{2^m \sqrt{2r^2 - 2ra_{m,0}}}{2^m} &= \frac{s}{2^m}\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ a_{m,n} &= \frac{2a_{m,n-1}^2}{r} - r, n \leq m \\ \sqrt{2r^2 - 2ra_{m,0}} &= \frac{s}{2^m}\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ a_{m,n} &= \frac{2a_{m,n-1}^2}{r} - r, n \leq m \\ \left(\sqrt{2r^2 - 2ra_{m,0}}\right)^2 &= \left(\frac{s}{2^m}\right)^2\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ a_{m,n} &= \frac{2a_{m,n-1}^2}{r} - r, n \leq m \\ 2r^2 - 2ra_{m,0} &= \left(\frac{s}{2^m}\right)^2\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ a_{m,n} &= \frac{2a_{m,n-1}^2}{r} - r, n \leq m \\ 2r^2 - 2ra_{m,0} &= \frac{s^2}{(2^m)^2}\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ a_{m,n} &= \frac{2a_{m,n-1}^2}{r} - r, n \leq m \\ 2r^2 - 2ra_{m,0} &= \frac{s^2}{2^{2m}}\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ a_{m,n} &= \frac{2a_{m,n-1}^2}{r} - r, n \leq m \\ 2r^2 - 2ra_{m,0} - 2r^2 &= \frac{s^2}{2^{2m}} - 2r^2\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ a_{m,n} &= \frac{2a_{m,n-1}^2}{r} - r, n \leq m \\ -2ra_{m,0} &= \frac{s^2}{2^{2m}} - 2r^2\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ a_{m,n} &= \frac{2a_{m,n-1}^2}{r} - r, n \leq m \\ \frac{-2ra_{m,0}}{-2r} &= \frac{\left(\frac{s^2}{2^{2m}}\right) - 2r^2}{-2r}\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ a_{m,n} &= \frac{2a_{m,n-1}^2}{r} - r, n \leq m \\ a_{m,0} &= \frac{\left(\frac{s^2}{2^{2m}}\right) - 2r^2}{-2r}\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ a_{m,n} &= \frac{2a_{m,n-1}^2}{r} - r, n \leq m \\ a_{m,0} &= \frac{\left(\frac{s^2}{2^{2m}}\right)}{-2r} - \frac{2r^2}{-2r}\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ a_{m,n} &= \frac{2a_{m,n-1}^2}{r} - r, n \leq m \\ a_{m,0} &= \frac{s^2}{2^{2m}(-2r)} - \frac{2r^2}{-2r}\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ a_{m,n} &= \frac{2a_{m,n-1}^2}{r} - r, n \leq m \\ a_{m,0} &= \frac{s^2}{-2^{2m+1}r} - \frac{2r^2}{-2r}\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ a_{m,n} &= \frac{2a_{m,n-1}^2}{r} - r, n \leq m \\ a_{m,0} &= \frac{s^2}{-2^{2m+1}r} + \frac{-2r^2}{-2r}\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ a_{m,n} &= \frac{2a_{m,n-1}^2}{r} - r, n \leq m \\ a_{m,0} &= \frac{s^2}{-2^{2m+1}r} + r\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ a_{m,n} &= \frac{2a_{m,n-1}^2}{r} - r, n \leq m \\ a_{m,0} &= r + \frac{s^2}{-2^{2m+1}r}\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ a_{m,n} &= \frac{2a_{m,n-1}^2}{r} - r, n \leq m \\ a_{m,0} &= r - \left(\frac{s^2}{-2^{2m+1}r}\right)\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ a_{m,n} &= \frac{2a_{m,n-1}^2}{r} - r, n \leq m \\ a_{m,0} &= r - \left(\frac{-s^2}{-2^{2m+1}r}\right)\end{aligned}$$

$$\begin{aligned}
\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\
a_{m,n} &= \frac{2a_{m,n-1}^2}{r} - r, n \leq m \\
a_{m,0} &= r - \frac{s^2}{2^{2m+1}r} \\
a_{m,n} &= \frac{2a_{m,n-1}^2}{r} - r, n \leq m \\
a_{m,0} &= r - \frac{s^2}{2^{2m+1}r} \\
r\hat{a}_{m,n} &= \frac{2(r\hat{a}_{m,n-1})^2}{r} - r, n \leq m \\
r\hat{a}_{m,0} &= r - \frac{s^2}{2^{2m+1}r} \\
r\hat{a}_{m,n} &= \frac{2r^2\hat{a}_{m,n-1}^2}{r} - r, n \leq m \\
r\hat{a}_{m,0} &= r - \frac{s^2}{2^{2m+1}r} \\
r\hat{a}_{m,n} &= 2r\hat{a}_{m,n-1}^2 - r, n \leq m \\
r\hat{a}_{m,0} &= r - \frac{s^2}{2^{2m+1}r} \\
\frac{r\hat{a}_{m,n}}{r} &= \frac{2r\hat{a}_{m,n-1}^2 - r}{r}, n \leq m \\
r\hat{a}_{m,0} &= r - \frac{s^2}{2^{2m+1}r} \\
\hat{a}_{m,n} &= \frac{2r\hat{a}_{m,n-1}^2 - r}{r}, n \leq m \\
r\hat{a}_{m,0} &= r - \frac{s^2}{2^{2m+1}r} \\
\hat{a}_{m,n} &= \frac{2r\hat{a}_{m,n-1}^2}{r} - \frac{r}{r}, n \leq m \\
r\hat{a}_{m,0} &= r - \frac{s^2}{2^{2m+1}r} \\
\hat{a}_{m,n} &= 2\hat{a}_{m,n-1}^2 - \frac{r}{r}, n \leq m \\
r\hat{a}_{m,0} &= r - \frac{s^2}{2^{2m+1}r} \\
\hat{a}_{m,n} &= 2\hat{a}_{m,n-1}^2 - 1, n \leq m \\
r\hat{a}_{m,0} &= r - \frac{s^2}{2^{2m+1}r} \\
\hat{a}_{m,n} &= 2\hat{a}_{m,n-1}^2 - 1, n \leq m \\
\frac{r\hat{a}_{m,0}}{r} &= \frac{r - \left(\frac{s^2}{2^{2m+1}r}\right)}{r} \\
\hat{a}_{m,n} &= 2\hat{a}_{m,n-1}^2 - 1, n \leq m \\
\hat{a}_{m,0} &= \frac{r - \left(\frac{s^2}{2^{2m+1}r}\right)}{r} \\
\hat{a}_{m,n} &= 2\hat{a}_{m,n-1}^2 - 1, n \leq m \\
\hat{a}_{m,0} &= \frac{r}{r} - \frac{\left(\frac{s^2}{2^{2m+1}r}\right)}{r}
\end{aligned}$$

$$\begin{aligned}
\hat{a}_{m,n} &= 2\hat{a}_{m,n-1}^2 - 1, n \leq m \\
\hat{a}_{m,0} &= 1 - \frac{\left(\frac{s^2}{2^{2m+1}r}\right)}{r} \\
\hat{a}_{m,n} &= 2\hat{a}_{m,n-1}^2 - 1, n \leq m \\
\hat{a}_{m,0} &= 1 - \frac{s^2}{2^{2m+1}r^2} \\
\hat{a}_{m,n} &= T_2^{\circ n}(\hat{a}_{m,0}), n \leq m \\
\hat{a}_{m,n} &= T_2^{\circ n}\left(1 - \frac{s^2}{2^{2m+1}r^2}\right), n \leq m \\
\hat{a}_{m,n} &= T_{2^n}\left(1 - \frac{s^2}{2^{2m+1}r^2}\right), n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{T_{2^n}^{(k)}(1)}{k!} \left(\left(1 - \frac{s^2}{2^{2m+1}r^2}\right) - 1\right)^k, n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{T_{2^n}^{(k)}(1)}{k!} \left(1 - \frac{s^2}{2^{2m+1}r^2} - 1\right)^k, n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{T_{2^n}^{(k)}(1)}{k!} \left(-\frac{s^2}{2^{2m+1}r^2}\right)^k, n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{2^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \left(-\frac{s^2}{2^{2m+1}r^2}\right)^k, n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{2^k \left(-\frac{s^2}{2^{2m+1}r^2}\right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!}, n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{\left(2\left(-\frac{s^2}{2^{2m+1}r^2}\right)\right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!}, n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{\left(-2\frac{s^2}{2^{2m+1}r^2}\right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!}, n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{\left(-\frac{2s^2}{2^{2m+1}r^2}\right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!}, n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{\left(-\frac{s^2}{2^{2m}r^2}\right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!}, n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{\left(-\frac{s^2}{(2^2)^m r^2}\right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!}, n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{\left(-\frac{s^2}{4^m r^2}\right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!}, n \leq m
\end{aligned}$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s^2}{4^m r^2} \right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!}, n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{\left(\frac{s^2}{r^2} \right)}{4^m} \right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!}, n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{\left(\frac{s}{r} \right)^2}{4^m} \right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!}, n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \frac{\left(\left(\frac{s}{r} \right)^2 \right)^k}{(4^m)^k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!}, n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \frac{\left(\frac{s}{r} \right)^{2k}}{(4^m)^k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!}, n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{\frac{(-1)^k \left(\frac{s}{r} \right)^{2k}}{(4^m)^k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!}, n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{\left(\frac{(-1)^k \left(\frac{s}{r} \right)^{2k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(4^m)^k} \right)}{(2k)!}, n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r} \right)^{2k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(4^m)^k (2k)!}, n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r} \right)^{2k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)! (4^m)^k}, n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r} \right)^{2k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \frac{1}{(4^m)^k}, n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r} \right)^{2k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \frac{1}{\prod_{j=0}^{k-1} 4^m}, n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r} \right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \frac{(2^n)^2 - j^2}{4^m}, n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r} \right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \frac{(2^n)^2 - j^2}{4^m}, n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r} \right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \frac{4^n - j^2}{4^m}, n \leq m$$

$$\begin{aligned}
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(\frac{4^n}{4^m} - \frac{j^2}{4^m} \right), n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(4^{n-m} - \frac{j^2}{4^m} \right), n \leq m \\
r\hat{a}_{m,n} &= r \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(4^{n-m} - \frac{j^2}{4^m} \right), n \leq m \\
a_{m,n} &= r \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(4^{n-m} - \frac{j^2}{4^m} \right), n \leq m \\
\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\
\cos(r, s) &= \lim_{m \rightarrow \infty} r \sum_{k=0}^{2^m} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(4^{m-m} - \frac{j^2}{4^m} \right) \\
\cos(r, s) &= \lim_{m \rightarrow \infty} r \sum_{k=0}^{2^m} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(4^0 - \frac{j^2}{4^m} \right) \\
\cos(r, s) &= \lim_{m \rightarrow \infty} r \sum_{k=0}^{2^m} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(1 - \frac{j^2}{4^m} \right) \\
\cos(r, s) &= r \lim_{m \rightarrow \infty} \sum_{k=0}^{2^m} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(1 - \frac{j^2}{4^m} \right) \\
\cos(r, s) &= r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!}
\end{aligned}$$

0.1.3.2.1 Partial Derivative

$$\begin{aligned}
\frac{\partial \cos(r, s)}{\partial s} &= \frac{\partial \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \right)}{\partial s} \\
\frac{\partial \cos(r, s)}{\partial s} &= \frac{\partial \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \frac{s^{2k}}{r^{2k}}}{(2k)!} \right)}{\partial s} \\
\frac{\partial \cos(r, s)}{\partial s} &= \frac{\partial \left(r \sum_{k=0}^{\infty} \frac{(-1)^k s^{2k}}{r^{2k} (2k)!} \right)}{\partial s} \\
\frac{\partial \cos(r, s)}{\partial s} &= \frac{\partial \left(\sum_{k=0}^{\infty} \frac{(-1)^k r s^{2k}}{r^{2k} (2k)!} \right)}{\partial s} \\
\frac{\partial \cos(r, s)}{\partial s} &= \frac{\partial \left(\sum_{k=0}^{\infty} \frac{(-1)^k s^{2k}}{r^{2k-1} (2k)!} \right)}{\partial s} \\
\frac{\partial \cos(r, s)}{\partial s} &= \sum_{k=0}^{\infty} \frac{(-1)^k \frac{\partial s^{2k}}{\partial s}}{r^{2k-1} (2k)!} \\
\frac{\partial \cos(r, s)}{\partial s} &= \sum_{k=1}^{\infty} \frac{(-1)^k (2k) s^{2k-1}}{r^{2k-1} (2k)!}
\end{aligned}$$

$$\begin{aligned}
\frac{\partial \cos(r, s)}{\partial s} &= \sum_{k=1}^{\infty} \frac{(-1)^k (2k) \frac{s^{2k-1}}{r^{2k-1}}}{(2k)!} \\
\frac{\partial \cos(r, s)}{\partial s} &= \sum_{k=1}^{\infty} \frac{(-1)^k (2k) \left(\frac{s}{r}\right)^{2k-1}}{(2k)!} \\
\frac{\partial \cos(r, s)}{\partial s} &= \sum_{k=1}^{\infty} \frac{(-1)^k (2k) \left(\frac{s}{r}\right)^{2k-1}}{(2k)(2k-1)!} \\
\frac{\partial \cos(r, s)}{\partial s} &= \sum_{k=1}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k-1}}{(2k-1)!} \\
\frac{\partial \cos(r, s)}{\partial s} &= \sum_{k=1}^{\infty} \frac{-(-1)^{k-1} \left(\frac{s}{r}\right)^{2k-1}}{(2k-1)!} \\
\frac{\partial \cos(r, s)}{\partial s} &= - \sum_{k=1}^{\infty} \frac{(-1)^{k-1} \left(\frac{s}{r}\right)^{2k-1}}{(2k-1)!} \\
\frac{\partial \cos(r, s)}{\partial s} &= - \sum_{k=1}^{\infty} \frac{(-1)^{k-1} \left(\frac{s}{r}\right)^{2(k-1)+1}}{(2k-1)!} \\
\frac{\partial \cos(r, s)}{\partial s} &= - \sum_{k=1}^{\infty} \frac{(-1)^{k-1} \left(\frac{s}{r}\right)^{2(k-1)+1}}{(2(k-1)+1)!} \\
\frac{\partial \cos(r, s)}{\partial s} &= - \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k+1}}{(2k+1)!} \\
\frac{\partial \cos(r, s)}{\partial s} &= \frac{-r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k+1}}{(2k+1)!}}{r} \\
\frac{\partial \cos(r, s)}{\partial s} &= \frac{-\sin(r, s)}{r}
\end{aligned}$$

0.1.3.3 Arcsine

$$\begin{aligned}
\arcsin: \{(r, y) \in \mathbb{R}^2 \mid r > 0, -r \leq y \leq r\} &\rightarrow \mathbb{R} \\
(r, y) &\mapsto s
\end{aligned}$$

$$\begin{aligned}
\arcsin(r, y) &= \frac{\arcsin(r, y) + \arcsin(r, y)}{2} \\
\arcsin(r, y) &= \frac{\arcsin(r, y) + \arcsin(r, y) + \arccos(r, y) - \arccos(r, y)}{2} \\
\arcsin(r, y) &= \frac{\arcsin(r, y) + \arcsin(r, -y) + \arccos(r, -y) - \arccos(r, y)}{2} \\
\arcsin(r, y) &= \frac{\arcsin(r, y) - \arcsin(r, y) + \arccos(r, -y) - \arccos(r, y)}{2} \\
\arcsin(r, y) &= \frac{\arccos(r, -y) - \arccos(r, y)}{2}
\end{aligned}$$

0.1.3.4 Sine

$$\sin: \{(r, s) \in \mathbb{R}^2 \mid r > 0\} \rightarrow \mathbb{R}$$

$$(r, s) \mapsto y$$

$$\sin^2(r, \alpha - \beta) + (r - \cos(r, \alpha - \beta))^2 = (\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2$$

$$\cos^2(r, \alpha - \beta) + \sin^2(r, \alpha - \beta) + (r - \cos(r, \alpha - \beta))^2 - \cos^2(r, \alpha - \beta) = (\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2$$

$$r^2 + (r - \cos(r, \alpha - \beta))^2 - \cos^2(r, \alpha - \beta) = (\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2$$

$$r^2 + r^2 - 2r \cos(r, \alpha - \beta) + \cos^2(r, \alpha - \beta) - \cos^2(r, \alpha - \beta) = (\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2$$

$$2r^2 - 2r \cos(r, \alpha - \beta) + \cos^2(r, \alpha - \beta) - \cos^2(r, \alpha - \beta) = (\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2$$

$$2r^2 - 2r \cos(r, \alpha - \beta) = (\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2$$

$$2r^2 - 2r \cos(r, \alpha - \beta) - 2r^2 = (\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2 - 2r^2$$

$$-2r \cos(r, \alpha - \beta) = (\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2 - 2r^2$$

$$\frac{-2r \cos(r, \alpha - \beta)}{-2r} = \frac{(\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2 - 2r^2}{-2r}$$

$$\cos(r, \alpha - \beta) = \frac{(\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2 - 2r^2}{-2r}$$

$$\cos(r, \alpha - \beta) = \frac{\cos^2(r, \alpha) - 2 \cos(r, \alpha) \cos(r, \beta) + \cos^2(r, \beta) + (\sin(r, \alpha) - \sin(r, \beta))^2 - 2r^2}{-2r}$$

$$\cos(r, \alpha - \beta) = \frac{\cos^2(r, \alpha) - 2 \cos(r, \alpha) \cos(r, \beta) + \cos^2(r, \beta) + \sin^2(r, \alpha) - 2 \sin(r, \alpha) \sin(r, \beta) + \sin^2(r, \beta) - 2r^2}{-2r}$$

$$\cos(r, \alpha - \beta) = \frac{(\cos^2(r, \alpha) + \sin^2(r, \alpha)) - 2 \cos(r, \alpha) \cos(r, \beta) + \cos^2(r, \beta) - 2 \sin(r, \alpha) \sin(r, \beta) + \sin^2(r, \beta) - 2r^2}{-2r}$$

$$\cos(r, \alpha - \beta) = \frac{(\cos^2(r, \alpha) + \sin^2(r, \alpha)) - 2 \cos(r, \alpha) \cos(r, \beta) + (\cos^2(r, \beta) + \sin^2(r, \beta)) - 2 \sin(r, \alpha) \sin(r, \beta) - 2r^2}{-2r}$$

$$\cos(r, \alpha - \beta) = \frac{r^2 - 2 \cos(r, \alpha) \cos(r, \beta) + (\cos^2(r, \beta) + \sin^2(r, \beta)) - 2 \sin(r, \alpha) \sin(r, \beta) - 2r^2}{-2r}$$

$$\cos(r, \alpha - \beta) = \frac{r^2 - 2 \cos(r, \alpha) \cos(r, \beta) + r^2 - 2 \sin(r, \alpha) \sin(r, \beta) - 2r^2}{-2r}$$

$$\cos(r, \alpha - \beta) = \frac{2r^2 - 2 \cos(r, \alpha) \cos(r, \beta) - 2 \sin(r, \alpha) \sin(r, \beta) - 2r^2}{-2r}$$

$$\cos(r, \alpha - \beta) = \frac{-2 \cos(r, \alpha) \cos(r, \beta) - 2 \sin(r, \alpha) \sin(r, \beta)}{-2r}$$

$$\cos(r, \alpha - \beta) = \frac{-2 (\cos(r, \alpha) \cos(r, \beta) + \sin(r, \alpha) \sin(r, \beta))}{-2r}$$

$$\cos(r, \alpha - \beta) = \frac{\cos(r, \alpha) \cos(r, \beta) + \sin(r, \alpha) \sin(r, \beta)}{r}$$

$$\cos(r, \alpha - \beta) = r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha - \beta}{r}\right)^{2k}}{(2k)!}$$

$$\cos(r, \alpha - \beta) = r \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{(-1)^k \left(\frac{\alpha - \beta}{r}\right)^{2k}}{(2k)!}$$

$$\cos(r, \alpha - \beta) = r \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{(-1)^k \sum_{j=0}^{2k} \binom{2k}{j} \left(\frac{\alpha}{r}\right)^{2k-j} \left(\frac{-\beta}{r}\right)^j}{(2k)!}$$

$$\begin{aligned}
\cos(r, \alpha - \beta) &= r \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{(-1)^k \sum_{j=0}^{2k} \frac{(2k)!}{j!(2k-j)!} \left(\frac{\alpha}{r}\right)^{2k-j} \left(\frac{-\beta}{r}\right)^j}{(2k)!} \\
\cos(r, \alpha - \beta) &= r \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{(-1)^k \sum_{j=0}^{2k} (2k)! \frac{(\frac{\alpha}{r})^{2k-j} (\frac{-\beta}{r})^j}{j!(2k-j)!}}{(2k)!} \\
\cos(r, \alpha - \beta) &= r \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{(-1)^k (2k)! \sum_{j=0}^{2k} \frac{(\frac{\alpha}{r})^{2k-j} (\frac{-\beta}{r})^j}{j!(2k-j)!}}{(2k)!} \\
\cos(r, \alpha - \beta) &= r \lim_{n \rightarrow \infty} \sum_{k=0}^n (-1)^k \sum_{j=0}^{2k} \frac{(\frac{\alpha}{r})^{2k-j} (\frac{-\beta}{r})^j}{j!(2k-j)!} \\
\cos(r, \alpha - \beta) &= r \lim_{n \rightarrow \infty} \left(\sum_{k=0}^n \frac{(-1)^k (\frac{\alpha}{r})^{2k}}{(2k)!} \sum_{j=0}^{n-k} \frac{(-1)^j (\frac{\beta}{r})^{2j}}{(2j)!} + \sum_{k=0}^{n-1} \frac{(-1)^k (\frac{\alpha}{r})^{2k+1}}{(2k+1)!} \sum_{j=0}^{n-1-k} \frac{(-1)^j (\frac{\beta}{r})^{2j+1}}{(2j+1)!} \right) \\
\cos(r, \alpha - \beta) &= r \left(\lim_{n \rightarrow \infty} \left(\sum_{k=0}^n \frac{(-1)^k (\frac{\alpha}{r})^{2k}}{(2k)!} \sum_{j=0}^{n-k} \frac{(-1)^j (\frac{\beta}{r})^{2j}}{(2j)!} \right) + \lim_{n \rightarrow \infty} \left(\sum_{k=0}^{n-1} \frac{(-1)^k (\frac{\alpha}{r})^{2k+1}}{(2k+1)!} \sum_{j=0}^{n-1-k} \frac{(-1)^j (\frac{\beta}{r})^{2j+1}}{(2j+1)!} \right) \right) \\
\cos(r, \alpha - \beta) &= r \left(\sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\alpha}{r})^{2k}}{(2k)!} \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\beta}{r})^{2k}}{(2k)!} + \lim_{n \rightarrow \infty} \left(\sum_{k=0}^{n-1} \frac{(-1)^k (\frac{\alpha}{r})^{2k+1}}{(2k+1)!} \sum_{j=0}^{n-1-k} \frac{(-1)^j (\frac{\beta}{r})^{2j+1}}{(2j+1)!} \right) \right) \\
\cos(r, \alpha - \beta) &= r \left(\sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\alpha}{r})^{2k}}{(2k)!} \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\beta}{r})^{2k}}{(2k)!} + \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\alpha}{r})^{2k+1}}{(2k+1)!} \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\beta}{r})^{2k+1}}{(2k+1)!} \right) \\
\cos(r, \alpha - \beta) &= r \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\alpha}{r})^{2k}}{(2k)!} \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\beta}{r})^{2k}}{(2k)!} + r \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\alpha}{r})^{2k+1}}{(2k+1)!} \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\beta}{r})^{2k+1}}{(2k+1)!} \\
\cos(r, \alpha - \beta) &= \frac{r \left(r \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\alpha}{r})^{2k}}{(2k)!} \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\beta}{r})^{2k}}{(2k)!} + r \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\alpha}{r})^{2k+1}}{(2k+1)!} \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\beta}{r})^{2k+1}}{(2k+1)!} \right)}{r} \\
\cos(r, \alpha - \beta) &= \frac{r^2 \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\alpha}{r})^{2k}}{(2k)!} \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\beta}{r})^{2k}}{(2k)!} + r^2 \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\alpha}{r})^{2k+1}}{(2k+1)!} \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\beta}{r})^{2k+1}}{(2k+1)!}}{r} \\
\cos(r, \alpha - \beta) &= \frac{\left(r \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\alpha}{r})^{2k}}{(2k)!} \right) \left(r \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\beta}{r})^{2k}}{(2k)!} \right) + r^2 \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\alpha}{r})^{2k+1}}{(2k+1)!} \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\beta}{r})^{2k+1}}{(2k+1)!}}{r} \\
\cos(r, \alpha - \beta) &= \frac{\left(r \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\alpha}{r})^{2k}}{(2k)!} \right) \left(r \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\beta}{r})^{2k}}{(2k)!} \right) + \left(r \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\alpha}{r})^{2k+1}}{(2k+1)!} \right) \left(r \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\beta}{r})^{2k+1}}{(2k+1)!} \right)}{r} \\
\cos(r, \alpha - \beta) &= \frac{\cos(r, \alpha) \cos(r, \beta) + \left(r \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\alpha}{r})^{2k+1}}{(2k+1)!} \right) \left(r \sum_{k=0}^{\infty} \frac{(-1)^k (\frac{\beta}{r})^{2k+1}}{(2k+1)!} \right)}{r}
\end{aligned}$$

$$\frac{\cos(r, \alpha) \cos(r, \beta) + \sin(r, \alpha) \sin(r, \beta)}{r} = \frac{\cos(r, \alpha) \cos(r, \beta) + \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r} \right)^{2k+1}}{(2k+1)!} \right) \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r} \right)^{2k+1}}{(2k+1)!} \right)}{r}$$

$$\sin(r, \alpha) \sin(r, \beta) = \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha}{r} \right)^{2k+1}}{(2k+1)!} \right) \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\beta}{r} \right)^{2k+1}}{(2k+1)!} \right)$$

$$\sin(r, s) \stackrel{\{\alpha, \beta\} \mapsto s}{=} r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r} \right)^{2k+1}}{(2k+1)!}$$

0.1.3.4.1 Partial Derivative

$$\frac{\partial \sin(r, s)}{\partial s} = \frac{\partial \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r} \right)^{2k+1}}{(2k+1)!} \right)}{\partial s}$$

$$\frac{\partial \sin(r, s)}{\partial s} = \frac{\partial \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \frac{s^{2k+1}}{r^{2k+1}}}{(2k+1)!} \right)}{\partial s}$$

$$\frac{\partial \sin(r, s)}{\partial s} = \frac{\partial \left(r \sum_{k=0}^{\infty} \frac{(-1)^k s^{2k+1}}{r^{2k+1} (2k+1)!} \right)}{\partial s}$$

$$\frac{\partial \sin(r, s)}{\partial s} = \frac{\partial \left(\sum_{k=0}^{\infty} \frac{(-1)^k r s^{2k+1}}{r^{2k+1} (2k+1)!} \right)}{\partial s}$$

$$\frac{\partial \sin(r, s)}{\partial s} = \frac{\partial \left(\sum_{k=0}^{\infty} \frac{(-1)^k s^{2k+1}}{r^{2k} (2k+1)!} \right)}{\partial s}$$

$$\frac{\partial \sin(r, s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k \frac{\partial s^{2k+1}}{\partial s}}{r^{2k} (2k+1)!}$$

$$\frac{\partial \sin(r, s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k (2k+1) s^{2k}}{r^{2k} (2k+1)!}$$

$$\frac{\partial \sin(r, s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k (2k+1) \frac{s^{2k}}{r^{2k}}}{(2k+1)!}$$

$$\frac{\partial \sin(r, s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k (2k+1) \left(\frac{s}{r} \right)^{2k}}{(2k+1)!}$$

$$\frac{\partial \sin(r, s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k (2k+1) \left(\frac{s}{r} \right)^{2k}}{(2k+1)(2k)!}$$

$$\frac{\partial \sin(r, s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r} \right)^{2k}}{(2k)!}$$

$$\frac{\partial \sin(r, s)}{\partial s} = \frac{r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r} \right)^{2k}}{(2k)!}}{r}$$

$$\frac{\partial \sin(r, s)}{\partial s} = \frac{\cos(r, s)}{r}$$

0.2 Circular Sector Sequence

0.2.1 Apothem

$$\begin{aligned}
A_{ORS_{n-1}} &= \frac{\overline{RS_{n-1}OX_n}}{2} \\
A_{ORS_{n-1}} &= \frac{\overline{OS_{n-1}RX_{n-1}}}{2} \\
\frac{\overline{RS_{n-1}OX_n}}{2} &= \frac{\overline{OS_{n-1}RX_{n-1}}}{2} \\
\overline{RS_{n-1}OX_n} &= \overline{OS_{n-1}RX_{n-1}} \\
\frac{\overline{RS_{n-1}OX_n}}{\overline{RS_{n-1}}} &= \frac{\overline{OS_{n-1}RX_{n-1}}}{\overline{RS_{n-1}}} \\
\overline{OX_n} &= \frac{\overline{OS_{n-1}RX_{n-1}}}{\overline{RS_{n-1}}} \\
\overline{OX_n} &= \frac{\overline{OS_{n-1}}\sqrt{\overline{OR}^2 - \overline{OX_{n-1}}^2}}{\overline{RS_{n-1}}} \\
\overline{OX_n} &= \frac{\overline{OS_{n-1}}\sqrt{\overline{OR}^2 - \overline{OX_{n-1}}^2}}{\sqrt{\overline{OR}^2 + \overline{OS_{n-1}}^2 - 2\overline{OS_{n-1}}\overline{OX_{n-1}}}}
\end{aligned}$$

0.2.2 Chord Length

$$\begin{aligned}
\overline{RS_n} &= \sqrt{\overline{RX_n}^2 + \overline{X_nS_n}^2} \\
\overline{RS_n} &= \sqrt{\overline{OR}^2 - \overline{OX_n}^2 + \overline{X_nS_n}^2} \\
\overline{RS_n} &= \sqrt{\overline{OR}^2 - \overline{OX_n}^2 + (\overline{OS_n} - \overline{OX_n})^2} \\
\overline{RS_n} &= \sqrt{\overline{OR}^2 - \overline{OX_n}^2 + \overline{OS_n}^2 - 2\overline{OS_n}\overline{OX_n} + \overline{OX_n}^2} \\
\overline{RS_n} &= \sqrt{\overline{OR}^2 + \overline{OS_n}^2 - 2\overline{OS_n}\overline{OX_n}}
\end{aligned}$$

0.2.3 Area

$$\begin{aligned}
A_0 &= \lim_{n \rightarrow \infty} 2^{n-1} A_{ORS_nS_{n-1}} \\
A_0 &= \lim_{n \rightarrow \infty} 2^{n-1} \frac{\overline{RS_{n-1}OS_n}}{2} \\
A_0 &= \frac{\lim_{n \rightarrow \infty} 2^{n-1} \overline{RS_{n-1}OS_n}}{2} \\
A_0 &= \frac{\lim_{n \rightarrow \infty} (2^{n-1} \overline{RS_{n-1}}) \overline{OS_n}}{2} \\
A_0 &= \frac{\lim_{n \rightarrow \infty} \overline{OS_n} (2^{n-1} \overline{RS_{n-1}})}{2} \\
A_0 &= \frac{\left(\lim_{n \rightarrow \infty} \overline{OS_n} \right) \left(\lim_{n \rightarrow \infty} 2^{n-1} \overline{RS_{n-1}} \right)}{2} \\
A_0 &= \frac{\overline{OR} \lim_{n \rightarrow \infty} 2^{n-1} \overline{RS_{n-1}}}{2} \\
A_0 &= \frac{\overline{OR} \text{arc } RS_0}{2}
\end{aligned}$$

0.3 Simple Harmonic Motion

$$[m] = \text{kg}$$

$$[\omega] = \text{m/s}$$

$$[k] = \text{N/m}$$

$$[r] = \text{m}$$

$$F = ma$$

$$F = -kx$$

$$ma = -kx$$

$$x(t) = \cos(r, \omega t)$$

$$a(t) = \frac{d^2 x}{dt^2}$$

$$a(t) = \frac{d^2 \cos(r, \omega t)}{dt^2}$$

$$a(t) = \frac{-\omega^2 \cos(r, \omega t)}{r^2}$$

$$a(x) = \frac{-\omega^2 x}{r^2}$$

$$m \frac{-\omega^2 x}{r^2} = -kx$$

$$\frac{m(-\omega^2 x)}{r^2} = -kx$$

$$\frac{-m\omega^2 x}{r^2} = -kx$$

$$\frac{-m\omega^2 x}{r^2} = \frac{-kr^2 x}{r^2}$$

$$m\omega^2 = kr^2$$

$$v(t) = \frac{dx}{dt}$$

$$v(t) = \frac{d \cos(r, \omega t)}{dt}$$

$$v(t) = \frac{-\omega \sin(r, \omega t)}{r}$$

$$K(t) = \frac{mv^2(t)}{2}$$

$$K(t) = \frac{m \left(\frac{-\omega \sin(r, \omega t)}{r} \right)^2}{2}$$

$$K(t) = \frac{m \frac{(-\omega \sin(r, \omega t))^2}{r^2}}{2}$$

$$K(t) = \frac{m \frac{(-\omega)^2 \sin^2(r, \omega t)}{r^2}}{2}$$

$$K(t) = \frac{m \frac{\omega^2 \sin^2(r, \omega t)}{r^2}}{2}$$

$$K(t) = \frac{\left(\frac{m\omega^2 \sin^2(r, \omega t)}{r^2} \right)}{2}$$

$$\begin{aligned}
K(t) &= \frac{\left(\frac{kr^2 \sin^2(r, \omega t)}{r^2}\right)}{2} \\
K(t) &= \frac{k \sin^2(r, \omega t)}{2} \\
U(t) &= \frac{kx^2(t)}{2} \\
U(t) &= \frac{k \cos^2(r, \omega t)}{2} \\
E &= K + U \\
E &= \frac{k \sin^2(r, \omega t)}{2} + \frac{k \cos^2(r, \omega t)}{2} \\
E &= \frac{k \sin^2(r, \omega t) + k \cos^2(r, \omega t)}{2} \\
E &= \frac{k(\sin^2(r, \omega t) + \cos^2(r, \omega t))}{2} \\
E &= \frac{kr^2}{2} \\
E &= \frac{m\omega^2}{2}
\end{aligned}$$

0.4 Uniform Circular Motion

$$\begin{aligned}
[m] &= \text{kg} \\
[\omega] &= \text{m/s} \\
[r] &= \text{m} \\
s &= \omega t \\
\omega &= \frac{\Delta s}{\Delta t} \\
\vec{r} &= (\vec{r}_x, \vec{r}_y) \\
\vec{r}(t) &= (\cos(r, \omega t), \sin(r, \omega t)) \\
\vec{v} &= \frac{d\vec{r}}{dt} \\
\vec{v}(t) &= \frac{d(\cos(r, \omega t), \sin(r, \omega t))}{dt} \\
\vec{v}(t) &= \left(\frac{d \cos(r, \omega t)}{dt}, \frac{d \sin(r, \omega t)}{dt} \right) \\
\vec{v}(t) &= \left(\frac{-\omega \sin(r, \omega t)}{r}, \frac{\omega \cos(r, \omega t)}{r} \right) \\
\vec{v}(t) &= \frac{\omega (-\sin(r, \omega t), \cos(r, \omega t))}{r} \\
v &= |\vec{v}| \\
v &= \left| \frac{\omega (-\sin(r, \omega t), \cos(r, \omega t))}{r} \right| \\
v &= \frac{|\omega (-\sin(r, \omega t), \cos(r, \omega t))|}{|r|} \\
v &= \frac{|\omega| |(-\sin(r, \omega t), \cos(r, \omega t))|}{|r|}
\end{aligned}$$

$$v = \frac{\omega |(-\sin(r, \omega t), \cos(r, \omega t))|}{|r|}$$

$$v = \frac{\omega \sqrt{(-\sin(r, \omega t))^2 + \cos^2(r, \omega t)}}{|r|}$$

$$v = \frac{\omega \sqrt{\sin^2(r, \omega t) + \cos^2(r, \omega t)}}{|r|}$$

$$v = \frac{\omega \sqrt{r^2}}{|r|}$$

$$v = \frac{\omega |r|}{|r|}$$

$$v = \omega$$

$$E = \frac{mv^2}{2}$$

$$E = \frac{m\omega^2}{2}$$

$$\vec{a} = \frac{d^2 \vec{r}}{dt^2}$$

$$\vec{a}(t) = \frac{d^2(\cos(r, \omega t), \sin(r, \omega t))}{dt^2}$$

$$\vec{a}(t) = \left(\frac{d^2 \cos(r, \omega t)}{dt^2}, \frac{d^2 \sin(r, \omega t)}{dt^2} \right)$$

$$\vec{a}(t) = \left(\frac{-\omega^2 \cos(r, \omega t)}{r^2}, \frac{-\omega^2 \sin(r, \omega t)}{r^2} \right)$$

$$\vec{a}(t) = \frac{-\omega^2 (\cos(r, \omega t), \sin(r, \omega t))}{r^2}$$

$$\vec{a} = \frac{-\omega^2 \vec{r}}{r^2}$$

$$a = |\vec{a}|$$

$$a = \left| \frac{-\omega^2 \vec{r}}{r^2} \right|$$

$$a = \frac{|-\omega^2 \vec{r}|}{|r^2|}$$

$$a = \frac{|-\omega^2| |\vec{r}|}{|r^2|}$$

$$a = \frac{\omega^2 |\vec{r}|}{|r^2|}$$

$$a = \frac{\omega^2 r}{|r^2|}$$

$$a = \frac{\omega^2 r}{r^2}$$

$$a = \frac{\omega^2}{r}$$

$$F = ma$$

$$F = m \frac{\omega^2}{r}$$

$$F = \frac{m\omega^2}{r}$$







